

PreACT 8/9 Printable Practice Exam 18

Math • Exam 05

35 questions • 40 minutes

PART A - QUESTIONS

M5-001

What is the value of $|-12| \div 3 + 2$?

- A) 2
- B) 4
- C) 6
- D) 10
- E) 14

M5-002

What value of x satisfies $8 - 2x = -10$?

- A) -9
- B) -1
- C) 1
- D) 9
- E) 18

M5-003

Three interior angles of a quadrilateral measure 85° , 95° , and 110° . What is the measure of the fourth interior angle?

- A) 60°
- B) 70°
- C) 80°
- D) 90°
- E) 110°

M5-004

An elevator is 120 meters above the ground when it begins descending at a constant rate. Its height is shown in the table.

t (seconds)	0	3	6	9
h (meters)	120	105	90	75

Which equation gives the elevator's height h after t seconds?

- A) $h = 120 - 5t$
- B) $h = 120 - 15t$
- C) $h = 120 + 5t$
- D) $h = 40 - 5t$
- E) $h = 5t$

M5-005

What is the range of the data set 4, 9, 11, 11, 18?

- A) 7
- B) 9
- C) 11

- D) 14
- E) 18

M5-006

What is $(4.5 \times 10^{-3})(2 \times 10^4)$ written in scientific notation?

- A) 9×10^{-7}
- B) 9×10^{-1}
- C) 9×10^1
- D) 9×10^7
- E) 90×10^1

M5-007

An online store discounts a \$120 item by 15% and then adds an \$8 shipping charge. What is the final cost before tax?

- A) \$94
- B) \$102
- C) \$110
- D) \$112
- E) \$128

M5-008

A tailor uses 7 meters of fabric to make 4 identical costumes. At the same rate, how many meters of fabric are needed to make 10 costumes?

- A) 13 meters
- B) 14 meters
- C) 17 meters
- D) 17.5 meters
- E) 28 meters

M5-009

A parallelogram has a base of 11 centimeters and a perpendicular height of 6 centimeters. What is its area?

- A) 17 square centimeters
- B) 33 square centimeters
- C) 66 square centimeters
- D) 72 square centimeters
- E) 121 square centimeters

M5-010

One letter is selected at random from the 7 letters in the word ALGEBRA. What is the probability that the selected letter is A?

- A) $\frac{1}{7}$
- B) $\frac{2}{7}$
- C) $\frac{1}{3}$
- D) $\frac{2}{5}$
- E) $\frac{1}{2}$

M5-011

What is the value of $\sqrt[3]{216} + \sqrt{49}$?

- A) 11

- B) 12
- C) 13
- D) 14
- E) 19

M5-012

The first three terms of an arithmetic sequence are -4 , 2 , and 8 . What is the twelfth term?

- A) 56
- B) 60
- C) 62
- D) 66
- E) 68

M5-013

An auditorium has 180 seats. Twenty-four seats are reserved for staff, and each visiting student group has 13 students. What is the greatest number of complete student groups that can be seated?

- A) 10
- B) 11
- C) 12
- D) 13
- E) 14

M5-014

Point V has coordinates $(-3, 4)$. A dilation centered at the origin with scale factor 2 is applied to V. What are the coordinates of the image of V?

- A) $(-6, 8)$
- B) $(-5, 6)$
- C) $(-1.5, 2)$
- D) $(3, -4)$
- E) $(6, -8)$

M5-015

The table shows whether 100 students buy lunch at school.

	Buys lunch	Does not buy lunch	Total
Grade 8	35	15	50
Grade 9	30	20	50
Total	65	35	100

One student is selected at random from the students who buy lunch. What is the probability that the selected student is in Grade 8?

- A) $\frac{7}{20}$
- B) $\frac{6}{13}$
- C) $\frac{7}{13}$
- D) $\frac{13}{20}$
- E) $\frac{7}{10}$

M5-016

A machine processes 450 grams of material in 3 minutes at a constant rate. How many kilograms of material does it process in 20 minutes?

- A) 0.3 kilograms
- B) 1.5 kilograms

- C) 3 kilograms
- D) 9 kilograms
- E) 30 kilograms

M5-017

A movie theater sells 28 tickets consisting only of adult and child tickets. Adult tickets cost \$9, child tickets cost \$6, and the total revenue is \$225. How many adult tickets are sold?

- A) 9
- B) 13
- C) 15
- D) 19
- E) 21

M5-018

A bicycle costs \$800 when new and loses 25% of its value each year. If this pattern continues, what is the bicycle's value after 2 years?

- A) \$400
- B) \$450
- C) \$500
- D) \$550
- E) \$600

M5-019

A cylindrical label covers only the curved side of a can with radius 4 centimeters and height 10 centimeters. What is the area of the label?

- A) 40π square centimeters
- B) 56π square centimeters
- C) 80π square centimeters
- D) 100π square centimeters
- E) 160π square centimeters

M5-020

The table shows the number of books read by 10 students during one month.

Books read	0	1	2	3
Students	2	4	3	1

What is the mean number of books read per student?

- A) 1.0
- B) 1.2
- C) 1.3
- D) 1.5
- E) 2.0

M5-021

What is the value of $(4^3)(4^{-1})$?

- A) $1/16$
- B) 4
- C) 8
- D) 16
- E) 64

M5-022

The equation $x^2 - 2x - 24 = 0$ has two solutions. What is the greater solution?

- A) -6
- B) -4
- C) 4
- D) 6
- E) 24

M5-023

The distance D , in kilometers, remaining on a trip after t hours is modeled by $D(t) = 150 - 12t$. After how many hours will 78 kilometers remain?

- A) 4 hours
- B) 5 hours
- C) 6 hours
- D) 7 hours
- E) 8 hours

M5-024

At the same time of day, a 1.5-meter post casts a 2-meter shadow and a tree casts a 14-meter shadow. Assuming the post and tree are vertical, how tall is the tree?

- A) 9.3 meters
- B) 10.5 meters
- C) 12 meters
- D) 15.5 meters
- E) 21 meters

M5-025

Service A charges $C = 12 + 3.5h$ dollars for h hours. Service B charges a starting fee plus a constant hourly amount, as shown in the table.

h (hours)	0	2	4
Service B cost (dollars)	20	26	32

For 8 hours, how does the cost of Service A compare with the cost of Service B?

- A) Service A costs \$4 less.
- B) Service A costs \$4 more.
- C) Service A costs \$8 less.
- D) Service A costs \$8 more.
- E) The two services cost the same amount.

M5-026

A rectangular photograph is 8 inches wide and 10 inches long. Both dimensions are increased by 25%. What is the perimeter of the enlarged photograph?

- A) 36 inches
- B) 40 inches
- C) 45 inches
- D) 50 inches
- E) 56.25 inches

M5-027

What value of x satisfies $2(3x - 4) + 5 = 4x + 15$?

- A) 4
- B) 6
- C) 8
- D) 9
- E) 11

M5-028

Which of the following numbers is irrational?

- A) $\sqrt[3]{64}$
- B) $\sqrt{121}$
- C) $-7/8$
- D) $0.333\ldots$
- E) $\sqrt{45}$

M5-029

What is the distance between the points $(1, -2)$ and $(7, 6)$ in the coordinate plane?

- A) 8
- B) 10
- C) 12
- D) 14
- E) 100

M5-030

A linear function f satisfies $f(-2) = 10$ and $f(4) = -2$. What is $f(1)$?

- A) -4
- B) 0
- C) 2
- D) 4
- E) 6

M5-031

Two fair number cubes, each with faces numbered 1 through 6, are rolled. What is the probability that the sum of the two results is 7?

- A) $1/12$
- B) $1/6$
- C) $1/4$
- D) $1/3$
- E) $7/12$

M5-032

A rectangular sandbox is 2.4 meters long and 1.5 meters wide. It is filled with sand to a uniform depth of 0.25 meter. Each delivery contains 0.15 cubic meter of sand. What is the least number of deliveries needed to fill the sandbox?

- A) 4
- B) 5
- C) 6
- D) 8
- E) 9

M5-033

Rental Plan A costs \$28 plus \$6 per hour. Rental Plan B costs \$52 plus \$3 per hour. For how many hours do the two plans cost the same amount?

- A) 4 hours
- B) 6 hours
- C) 8 hours
- D) 10 hours
- E) 12 hours

M5-034

A rectangular sheet of cardboard measures 18 centimeters by 14 centimeters. Squares with side length 2 centimeters are cut from all four corners, and the remaining cardboard is folded to form an open box. What is the volume of the box?

- A) 140 cubic centimeters
- B) 196 cubic centimeters
- C) 252 cubic centimeters
- D) 280 cubic centimeters
- E) 504 cubic centimeters

M5-035

The mean of 6 daily measurements is 14. One measurement, 20, is removed. What is the mean of the remaining 5 measurements?

- A) 10.8
- B) 12.0
- C) 12.8
- D) 13.2
- E) 16.8

PART B - ANSWER KEY + EXPLANATIONS

M5-001 - Correct: C) 6

Explanation: $|-12| = 12$. Then $12 \div 3 + 2 = 4 + 2 = 6$.

M5-002 - Correct: D) 9

Explanation: Subtract 8 from both sides to get $-2x = -18$, and then divide by -2 to get $x = 9$.

M5-003 - Correct: B) 70°

Explanation: The interior angles of a quadrilateral sum to 360° . The fourth angle is $360^\circ - 85^\circ - 95^\circ - 110^\circ = 70^\circ$.

M5-004 - Correct: A) $h = 120 - 5t$

Explanation: The elevator begins at 120 meters and descends 15 meters every 3 seconds, a rate of -5 meters per second. Therefore, $h = 120 - 5t$.

M5-005 - Correct: D) 14

Explanation: The range is the greatest value minus the least value: $18 - 4 = 14$.

M5-006 - Correct: C) 9×10^1

Explanation: Multiply the coefficients and add the exponents: $4.5 \times 2 = 9$ and $10^{-3} \times 10^4 = 10^1$. Therefore, the product is 9×10^1 .

M5-007 - Correct: C) \$110

Explanation: The discounted price is $\$120 \times 0.85 = \102 . Adding \$8 shipping gives $\$102 + \$8 = \$110$.

M5-008 - Correct: D) 17.5 meters

Explanation: The fabric rate is $7 \div 4 = 1.75$ meters per costume. For 10 costumes, $10 \times 1.75 = 17.5$ meters.

M5-009 - Correct: C) 66 square centimeters

Explanation: A parallelogram's area is base $\times$ perpendicular height, so the area is $11 \times 6 = 66$ square centimeters.

M5-010 - Correct: B) $2/7$

Explanation: The word ALGEBRA has 7 letters, and 2 of them are A. The probability is $2/7$.

M5-011 - Correct: C) 13

Explanation: $\sqrt[3]{216} = 6$ and $\sqrt{49} = 7$, so $\sqrt[3]{216} + \sqrt{49} = 6 + 7 = 13$.

M5-012 - Correct: C) 62

Explanation: The common difference is 6. There are 11 increases from the first term to the twelfth, so the twelfth term is $-4 + 11 \times 6 = 62$.

M5-013 - Correct: C) 12

Explanation: After reserving 24 seats, $180 - 24 = 156$ seats remain. Since $156 \div 13 = 12$, at most 12 complete groups can be seated.

M5-014 - Correct: A) $(-6, 8)$

Explanation: A dilation centered at the origin with scale factor 2 multiplies both coordinates by 2, so $(-3, 4)$ maps to $(-6, 8)$.

M5-015 - Correct: C) $7/13$

Explanation: Of the 65 students who buy lunch, 35 are in Grade 8. The probability is $35/65 = 7/13$.

M5-016 - Correct: C) 3 kilograms

Explanation: The machine processes $450 \div 3 = 150$ grams per minute. In 20 minutes it processes $150 \times 20 = 3,000$ grams, or 3 kilograms.

M5-017 - Correct: D) 19

Explanation: Let a be adult tickets and c be child tickets. Then $a + c = 28$ and $9a + 6c = 225$. Substituting $c = 28 - a$ gives $3a + 168 = 225$, so $a = 19$.

M5-018 - Correct: B) \$450

Explanation: Losing 25% leaves 75% of the value each year. After 2 years, the value is $\$800 \times 0.75^2 = \$800 \times 0.5625 = \$450$.

M5-019 - Correct: C) 80π square centimeters

Explanation: The curved side area of a cylinder is $2\pi rh$. With $r = 4$ and $h = 10$, the area is $2 \times \pi \times 4 \times 10 = 80\pi$ square centimeters.

M5-020 - Correct: C) 1.3

Explanation: The total books read is $0(2) + 1(4) + 2(3) + 3(1) = 13$. Dividing by 10 students gives a mean of $13/10 = 1.3$.

M5-021 - Correct: D) 16

Explanation: Multiply powers with the same base by adding exponents: $(4^3)(4^{-1}) = 4^2 = 16$.

M5-022 - Correct: D) 6

Explanation: Factoring gives $x^2 - 2x - 24 = (x - 6)(x + 4)$, so the solutions are 6 and -4 . The greater solution is 6.

M5-023 - Correct: C) 6 hours

Explanation: Set $150 - 12t = 78$. Then $-12t = -72$, so $t = 6$ hours.

M5-024 - Correct: B) 10.5 meters

Explanation: The post and tree form similar triangles with their shadows. The height-to-shadow ratio is $1.5/2$, so the tree's height is $14 \times (1.5/2) = 10.5$ meters.

M5-025 - Correct: A) Service A costs \$4 less.

Explanation: For 8 hours, Service A costs $12 + 3.5(8) = \$40$. Service B has a rate of \$3 per hour and a starting fee of \$20, so it costs $20 + 3(8) = \$44$. Service A costs \$4 less.

M5-026 - Correct: C) 45 inches

Explanation: Increasing both dimensions by 25% gives dimensions $8 \times 1.25 = 10$ inches and $10 \times 1.25 = 12.5$ inches. The perimeter is $2(10 + 12.5) = 45$ inches.

M5-027 - Correct: D) 9

Explanation: Expanding gives $6x - 8 + 5 = 4x + 15$. Thus, $6x - 3 = 4x + 15$, so $2x = 18$ and $x = 9$.

M5-028 - Correct: E) $\sqrt{45}$

Explanation: $\sqrt[3]{64} = 4$, $\sqrt{121} = 11$, $-7/8$ is a ratio of integers, and $0.333\ldots$ is a repeating decimal, so those four numbers are rational. Since 45 is not a perfect square, $\sqrt{45}$ is irrational.

M5-029 - Correct: B) 10

Explanation: The horizontal change is $7 - 1 = 6$, and the vertical change is $6 - (-2) = 8$. The distance is $\sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$.

M5-030 - Correct: D) 4

Explanation: The slope is $(-2 - 10)/(4 - (-2)) = -12/6 = -2$. The function is $f(x) = -2x + 6$, so $f(1) = 4$.

M5-031 - Correct: B) $1/6$

Explanation: There are 36 equally likely ordered outcomes. The 6 outcomes with sum 7 are (1, 6), (2, 5), (3, 4), (4, 3), (5, 2), and (6, 1), so the probability is $6/36 = 1/6$.

M5-032 - Correct: C) 6

Explanation: The sandbox volume is $2.4 \times 1.5 \times 0.25 = 0.9$ cubic meter. Each delivery contains 0.15 cubic meter, so $0.9 \div 0.15 = 6$ deliveries are needed.

M5-033 - Correct: C) 8 hours

Explanation: Set the plan costs equal: $28 + 6h = 52 + 3h$. Then $3h = 24$, so $h = 8$ hours.

M5-034 - Correct: D) 280 cubic centimeters

Explanation: After the corner squares are removed, the box base is $(18 - 4)$ by $(14 - 4)$, or 14 centimeters by 10 centimeters, and its height is 2 centimeters. The volume is $14 \times 10 \times 2 = 280$ cubic centimeters.

M5-035 - Correct: C) 12.8

Explanation: The total of the 6 measurements is $6 \times 14 = 84$. Removing 20 leaves 64, and $64 \div 5 = 12.8$.